

WasteSmart

ABSTRACT

Waste management refers to the process of collecting, treating, and disposing of waste materials in a safe and environmentally responsible manner. It involves various activities aimed at minimizing the negative impacts of waste on human health, the environment, and society as a whole. Waste collection involves the gathering and transportation of waste from its source to a designated disposal facility. This can be done through curbside collection by municipal waste management services or private waste collection companies. In this application the pickup module has collecting the waste materials. The project aims to create an efficient and sustainable waste management solution by incorporating various modules to streamline operations, enhance user experience, and promote recycling. With a focus on collaboration between administrators, pickup agents, workers, users, and recycling units, our system provides an integrated platform for effective waste management.

The Admin module serves as the central control hub of the waste management system. Administrators can securely log in, manage their account credentials, and access various administrative functions. These functions include viewing and verifying recycle units, managing waste types and their rates, handling worker notifications and verification, allocating areas to workers, managing worker categories, and addressing user complaints and feedback.

The Pickup module is designed for pickup agents responsible for collecting waste from users. Pickup agents can sign up, log in, and manage their profiles. They can view user waste requests, collect waste from designated locations, and update the status of the requests. Pickup agents can also change their account passwords for enhanced security.

The Worker module caters to the needs of workers involved in waste management. Workers can sign up, log in, and view their profiles. They can access user requests, update the status of those requests, and view the areas allocated to them. The module allows workers to efficiently manage their tasks and ensure timely waste collection.

The User module focuses on providing a user-friendly interface for individuals who generate waste and wish to contribute to proper waste management. Users can sign up, log in, and manage their profiles. They can add waste requests for pickup, view payment details, and track the status of their requests. Users can also submit complaints, view the replies received, provide feedback, explore worker categories, view recycling products, add products to their cart, and make payments. Additionally, users can view worker notifications and apply for relevant positions.

The Recycle Unit module caters to recycling facilities responsible for processing and recycling waste. Recycle units can sign up, log in, and manage their profiles. They can view the status of collected user requests, manage recycling products, view the status of product orders, and handle payment transactions.

Existing System: Traditional Waste Management

The current waste management system relies heavily on manual processes and lacks efficient integration and user engagement. Here are the key characteristics and limitations of the existing system:

Key Characteristics:

1. Manual Waste Collection:

- Waste collection is often done manually by municipal or private waste management services.
- Limited coordination between collection agents and users.

2. Limited Recycling:

- Recycling practices are not widely adopted or integrated.
- Lack of dedicated platforms for managing recycling units and products.

3. Inefficient Communication:

- Users have limited means to request waste collection or provide feedback.
- Communication between users, workers, and administrators is often fragmented.

4. Minimal User Engagement:

- Users have limited interaction with the waste management system.
- Lack of incentives or educational resources to promote responsible waste disposal and recycling.

5. Basic Administrative Control:

- Administrators handle tasks such as waste type management and worker allocation manually.
- Limited tools for monitoring and optimizing waste collection routes and schedules.

6. Inconsistent Data Management:

- Data on waste collection, recycling, and user feedback is often not centralized.
- Difficulty in tracking and analyzing waste management performance.

Proposed System: WasteSmart

The proposed system, WasteSmart, leverages modern technology to create a comprehensive, efficient, and sustainable waste management solution. Here are the key features and benefits of the proposed system:

Key Features and Benefits:

1. Admin Module:

- **Central Control Hub:**
 - Secure login and account management for administrators.
 - View and verify recycle units, manage waste types and rates.
 - Handle worker notifications and verification, allocate areas to workers.
 - Address user complaints and feedback.
- **Data-Driven Management:**

- Centralized data on waste collection, recycling, and user feedback.
- Tools for monitoring and optimizing waste collection routes and schedules.

2. Pickup Module:

- **Agent Management:**
 - Pickup agents can sign up, log in, and manage their profiles.
 - View user waste requests, collect waste from designated locations.
 - Update request status and change account passwords for security.
- **Efficient Operations:**
 - Streamlined waste collection process with real-time updates on request status.

3. Worker Module:

- **Worker Interface:**
 - Workers can sign up, log in, and view their profiles.
 - Access user requests, update request status, and view allocated areas.
- **Task Management:**
 - Efficient management of tasks to ensure timely waste collection.

4. User Module:

- **User-Friendly Interface:**
 - Users can sign up, log in, and manage their profiles.
 - Add waste requests for pickup, view payment details, track request status.
- **Enhanced Engagement:**
 - Submit complaints, view replies, provide feedback.
 - Explore worker categories, view recycling products, add products to cart, and make payments.
 - View worker notifications and apply for relevant positions.

Signup/Login

Profile Management

Waste Pickup Request & Payment Viewing

Complaints & Feedback Submission

View Worker Categories

Embedded-Based Waste Classification Feedback:

- **While the user does not manually classify waste, the system uses embedded hardware (e.g., sensors, cameras, microcontrollers) in smart bins to:**
 - **Automatically detect and sort waste (organic, plastic, metal, etc.)**
 - **Notify users if waste is incorrectly placed.**
- **Users can receive alerts or suggestions based on classification results from the bin's embedded system.**

Notifications

5. Recycling Store: Browse Products, Add to Cart, Make Payment

6. Apply for Worker Roles

7. Recycle Unit Module:

- **Recycling Facility Management:**
 - Recycle units can sign up, log in, and manage their profiles.
 - View the status of collected user requests, manage recycling products.
 - Handle product orders and payment transactions.
- **Promoting Recycling:**
 - Integrated platform for recycling facilities to process and manage recycled materials.

Proposed System Benefits:

1. Integrated Platform:

- A unified system for administrators, pickup agents, workers, users, and recycling units.
- Seamless coordination and communication between all stakeholders.

2. Efficient Waste Collection:

- Real-time updates and streamlined processes for waste collection.
- Optimized routes and schedules for pickup agents.

3. Enhanced User Experience:

- User-friendly interfaces for easy request management and feedback.
- Incentives and educational resources to promote responsible waste disposal and recycling.

4. Sustainable Practices:

- Promotes recycling through dedicated modules for recycling facilities.
- Encourages users to participate in recycling initiatives.

5. **Data-Driven Decision Making:**

- Centralized data for monitoring and analyzing waste management performance.
- Tools for administrators to optimize operations and improve efficiency.

By integrating these features, WasteSmart aims to revolutionize waste management practices, enhance user engagement, and contribute to a cleaner and more sustainable environment.

MODULE DESCRIPTION

Our Proposed system consists of following modules:

Admin

- Login
- Change Password
- View recycle unit and verify
- Manage type of waste and its rate
- Manage Worker Notification and verify
- Area allocation to worker
- View Users
- Worker category management
- View complaints and send reply
- View feedback

Pickup

- Signup
- Login
- View Profile
- View user waste request and collect
- Change password

Worker

- Signup
- Login
- View profile
- View user request and update status
- View allocated area

User

- Signup
- Login
- Manage profile
- Add waste request and view payment
- Send complaint and view reply
- Manage feedback
- View worker category
- User view recycling product, add to cart and make payment
- View worker notification and apply

Recycle Unit

- Signup
- Login
- Manage Profile
- View collected user request status
- Manage recycling product
- View product order status and payment

REQUIREMENTS

PROCESSOR	-	INTEL PENTIUM IV	HARD DISK SPACE	-	320 GB
DISPLAY	-	15" COLOR MONITOR	MEMORY	-	4GB
FLOPPY DISK DRIVE	-	1.44 MB			

KEYBOARD - 104 KEYS

COMPONENTS - MOUSE ATTACHED

Software Requirements

OPERATING SYSTEM - WINDOWS 7

WEB - Python/DJANGO/HTML/CSS //

APPLICATION - FLUTTER/Android (Java)